2023-2024

S6 MATH CP

PAPER II

Name: _____

Class: _____( )

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Six Final Examination

MATHEMATICS Compulsory Part
PAPER 2

24/01/2024

11:05 am – 12:20 pm (1¼ hours)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first write your Name Class and Class Number and insert the information required in the spaces provided. No extra time will be given for writing your Name, Class and Class number after the 'Time is up' announcement.

2. When told to open this book, you should check that all questions are there. Look for the words ' **END OF PAPER**' after the last question.

3. All questions carry equal marks.

4. **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answer clearly; otherwise you will lose marks if the answers cannot be captured.

5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

6. No marks will be deducted for wrong answers.

There are 30 questions in Section A and 15 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.

Choose the best answer for each question.

Section A

1. $\dfrac{(6x^2)^2}{3(x^{-2})^{-1}}$ =

   A. $4x^2$.
   B. $4x^4$.
   C. $12x^2$.
   D. $12x^4$.

2. If $s = \dfrac{u+v}{u-v}$, then u =

   A. $\dfrac{s-1}{v(s+1)}$.
   B. $\dfrac{v(s-1)}{s+1}$.
   C. $\dfrac{s+1}{v(s-1)}$.
   D. $\dfrac{v(s+1)}{s-1}$.

3. $25 - (a - 2b)^2$ =

   A. $(5 + a - 2b)(5 - a - 2b)$.
   B. $(5 + a + 2b)(5 - a + 2b)$.
   C. $(5 + a + 2b)(5 - a - 2b)$.
   D. $(5 + a - 2b)(5 - a + 2b)$.

4. If $x = 2.34$ is correct to the nearest 0.02, then

   A. $2.33 < x \leq 2.35$.
   B. $2.33 \leq x < 2.35$.
   C. $2.32 < x \leq 2.36$.
   D. $2.32 \leq x < 2.36$.

5. The solution of $7 + 2x \le -3$ or $\frac{1-x}{2} \ge 2$ is

A. $x \le -5$ or $x \ge -3$.

B. $-5 \le x \le -3$.

C. $x \le -3$.

D. $x \le -5$.

6. $\frac{4}{2m+5} - \frac{3}{2m-5} =$

A. $\frac{8m-35}{4m^2-25}$.

B. $\frac{8m-21}{4m^2-25}$.

C. $\frac{2m-35}{4m^2-25}$.

D. $\frac{2m-21}{4m^2-25}$.

7. If A and B are constants such that $(x + A)(x - 4) + 3A \equiv x^2 + (3 - B)x - 1$, then $B =$

A. 6.

B. 3.

C. –2.

D. –6.

8. Let $f(x) = x^3 + kx^2 - kx - 9$, where k is a constant. If $x + k$ is a factor of $f(x)$, then $k =$

A. 1.

B. 2.

C. 0 or 9.

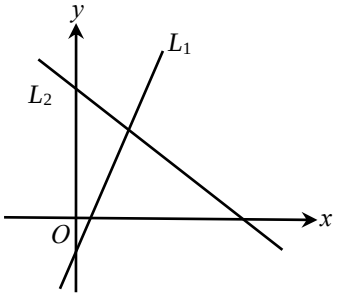
D. –3 or 3.

9. In the figure, the equations of the straight lines L_1 and L_2 are $3x + ay - b = 0$ and $cx + y + d = 0$ respectively, where a, b, c and d are constants. Which of the following must be true?

I. $a < 0$

II. $d > 0$

III. $bc + 3d < 0$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

10. It is given that $a : b = 5 : 2$ and $a : c = 2 : 3$. If $b = 12$, then $a + c =$

A. 15.
B. 30.
C. 60.
D. 75.

11. The scale of a map is 1 : 2000. If the area of a carpark on the map is 40 cm^2, then the actual area of the carpark is

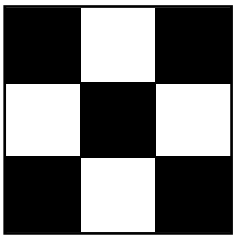
A. 8 km^2.
B. 0.8 km^2.
C. 0.16 km^2.
D. 0.016 km^2.

12. Let $p(x)$ be a polynomial which is divisible by $x - 3$. When $p(x)$ is divided by $x - 1$, the remainder is -2. Find the remainder when $p(x)$ is divided by $(x - 3)(x - 1)$.

A. $x - 3$
B. $x + 3$
C. $3x - 1$
D. $3x + 1$

13. The figure below consists of nine identical squares and five of them are shaded. The number of axes of reflectional symmetry of the figure is

A. 2.
B. 4.
C. 5.
D. 9.



14. Suppose z varies directly as the cube of x and inversely as the square root of y. Which of the following must be constant?

A. $\frac{x^3}{yz^2}$
B. $\frac{yz^2}{x^6}$
C. $\frac{y^2z}{x^3}$
D. $\frac{z}{y^2x^3}$

15. The base radius of a solid right circular cone is 12 cm. If the volume of the circular cone is 240π cm^3, then the total surface area of the circular cone is

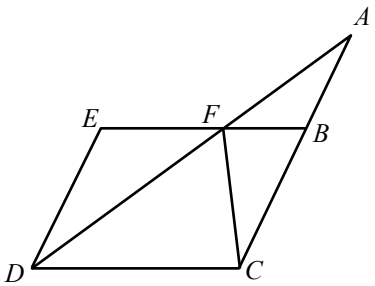
A. 156π cm^2.
B. 180π cm^2.
C. 300π cm^2.
D. 324π cm^2.

16. A sum of \$5000 is deposited at an interest rate of 4% per annum for 5 years, compounded half-yearly. Find the interest correct to the nearest dollar.

A. \$1000
B. \$1083
C. \$1095
D. \$1101

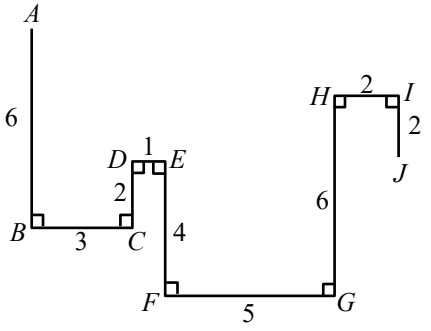
17. In the figure, $BCDE$ is a parallelogram. F is a point on BE such that DF produced meets CB produced at A and $EF : FB = 3 : 2$. If the area of $\triangle ABF$ is 8 cm^2, then the area of $\triangle CDF$ is

A. 40 cm^2.
B. 36 cm^2.
C. 32 cm^2.
D. 30 cm^2.



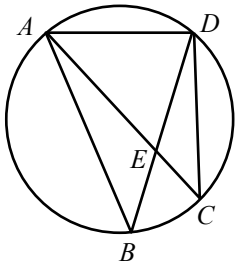
18. In the figure, the shortest distance between A and J is

A. $\sqrt{137}$.
B. $\sqrt{129}$.
C. $\sqrt{113}$.
D. $\sqrt{97}$.



19. In the figure, $ABCD$ is a circle. AC and BD intersect at E. If $\angle ACD = 40°$, $\angle BDC = 20°$ and $\overset{\frown}{AB} : \overset{\frown}{CD} = 3 : 2$, then $\angle DAB =$

A. 70°.
B. 68°.
C. 65°.
D. 62°.

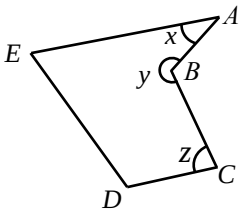


20. The figure shows a pentagon $ABCDE$ with AE // CD. Which of the following must be true?

I. The sum of the interior angles of $ABCDE$ is 540°.

II. $y - x - z = 0°$

III. $x + y + z = 360°$



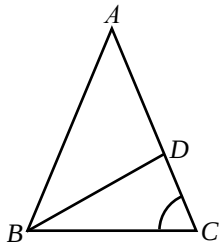
A. I and II only

B. I and III only

C. II and III only

D. I, II and III

21. In the figure, $\triangle ABC$ is an isosceles triangle with $AB = AC$. D is a point on AC such that $AD = DB = BC$. Find $\angle ACB$.



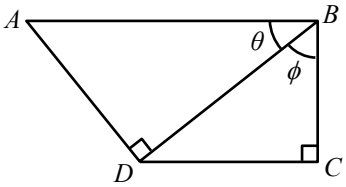
A. 72°

B. 76°

C. 82°

D. 86°

22. In the figure, $\angle ADB = \angle BCD = 90°$, $\angle ABD = \theta$ and $\angle CBD = \phi$. If $AB = p$, then $CD =$



A. $p \cos \theta \sin \phi$.

B. $p \sin \theta \cos \phi$.

C. $\dfrac{p \cos \phi}{\sin \theta}$.

D. $\dfrac{p \cos \theta}{\sin \phi}$.

23. For $0° \le \theta \le 90°$, the greatest value of $\dfrac{3 - 2\sin^2 \theta}{3\cos^2(90° - \theta) + 2\cos^2 \theta}$ is

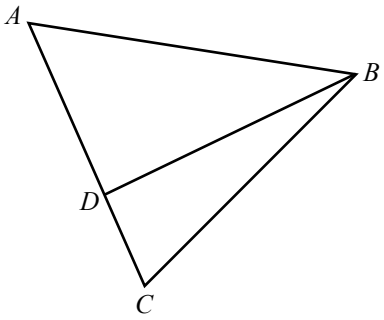
A. $\dfrac{1}{2}$.

B. 1.

C. $\dfrac{3}{2}$.

D. $\dfrac{1}{3}$.

24. The figure shows $\triangle ABC$. D is a point on AC such that $AD : DC = 9 : 5$. If $AC = 42$ cm, $BD = 36$ cm and $AB = 45$ cm, then the perimeter of $\triangle ABC$ is



A. 87 cm.
B. 126 cm.
C. 132 cm.
D. 144 cm.

25. The point P is rotated clockwise about the origin through 90° to the point Q. Q is then translated downwards by 4 units to the point R. If the coordinates of the point R are $(1, -3)$, then the polar coordinates of P are

A. $(1, 135°)$.
B. $(1, 315°)$.
C. $(\sqrt{2}, 135°)$.
D. $(\sqrt{2}, 315°)$.

26. The equation of the circle C is $2x^2 + 2y^2 - 6x + 12y - 135 = 0$. Which of the following about C is true?

A. The origin lies inside C.
B. The area of C is greater than 250.
C. The entire circle lies in the fourth quadrant.
D. The coordinates of the centre of C are $(3, -6)$.

27. Let $A(3, 5)$ and $B(10, -6)$ be two points in a rectangular coordinate plane. If P is a moving point in the same coordinate plane such that $AP = AB$, then the locus of P is

A. a circle.
B. a straight line.
C. two straight lines.
D. a parabola.

28. Two numbers are randomly drawn at the same time from the six cards numbered 1, 2, 3, 4, 5 and 6 respectively. Find the probability that the product of the numbers drawn is an even number.

A. $\frac{1}{2}$
B. $\frac{1}{5}$
C. $\frac{4}{5}$
D. $\frac{5}{6}$

29. The mean of 30 integers is 24. If the mean of 10 of these 30 integers is 26, then the mean of the other 20 integers is

A. 20.

B. 23.

C. 24.

D. 32.

30. Consider the following positive integers:

3 4 5 5 5 5 6 8 11 p q

If the range of the above integers is 8, which of the following must be true?

I. The mean of the integers is 5.

II. The median of the integers is 5.

III. The mode of the integers is 5.

A. I only

B. II only

C. I and III only

D. II and III only

Section B

31. The L.C.M. of $2x^3y^2z^6$, $3x^2y^5z$ and $9xy^2z^8$ is

A. xy^2z.

B. $x^2y^2z^6$.

C. $18x^3y^5z^8$.

D. $18xy^2z^6$.

32. It is given that $\log_8 y$ is a linear function of $\log_4 x$. If the slope and the intercept on the horizontal axis of the graph of the linear function are 5 and 2 respectively, then

A. $y^2 = 2^{12}x^{15}$.

B. $y^2 = \dfrac{x^{15}}{2^{60}}$.

C. $x^{15}y^2 = 2^{12}$.

D. $x^{15}y^2 = 2^{60}$.

33. $101000000011001_2 =$

A. $5 \times 2^{12} + 25$.

B. $5 \times 2^{12} + 50$.

C. $5 \times 2^{13} + 25$.

D. $5 \times 2^{13} + 50$.

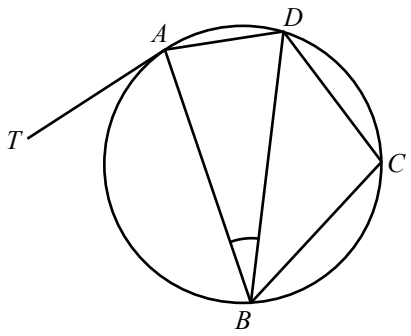
34. In the figure, TA is the tangent to the circle $ABCD$ at A. If $\angle TAB = 74°$ and $\angle BCD = 98°$, then $\angle ABD =$

A. 32°.

B. 17°.

C. 22°.

D. 24°.



35. Which of the following is the greatest?

A. 128^{812}

B. 218^{821}

C. 812^{281}

D. 821^{128}

36. If k is a real number, then the real part of $\dfrac{6-i^5}{2i^4+ki^3}$ is

A. $\dfrac{-2+6k}{4-k^2}$.

B. $\dfrac{12+k}{4-k^2}$.

C. $\dfrac{-2+6k}{4+k^2}$.

D. $\dfrac{12+k}{4+k^2}$.

37. Consider the following system of inequalities:

$$\begin{cases} x+2y \le 40 \\ x+y \ge 26 \\ y \ge x-10 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the constant k such that the least value of $2x - y + k$ is 13, where (x, y) is a point lying in D.

A. 3

B. 10

C. –15

D. –17

38. If a, b, c, d is an arithmetic sequence, which of the following must be true?

I. $a < b < c < d$

II. $c = \frac{1}{3}a + \frac{2}{3}d$

III. 2^a, 2^b, 2^c, 2^d is a geometric sequence.

A. I only

B. II only

C. I and III only

D. II and III only

39. Find the range of values of the constant k such that the circle $x^2 + y^2 - 12x - 24y - 320 = 0$ and the straight line $x - 2y + k = 0$ intersect.

A. $-32 \le k \le 68$

B. $-32 < k < 68$

C. $k \le -68$ or $k \ge 32$

D. $k < -68$ or $k > 32$

40. For $0° \le \theta < 360°$, how many roots does the equation $2\cos^2\theta + 3\sin\theta = 3$ have?

A. 1

B. 2

C. 3

D. 4

41. Let O be the origin. The coordinates of the point A are (48, 36). If the coordinates of the orthocentre of $\triangle OAB$ are (36, 12), then the x-coordinate of B is

A. –36.

B. –12.

C. 72.

D. 96.

42. A row is formed by 6 girls and 5 boys. If the boys and girls stand alternately, how many different rows can be formed?

A. 7280

B. 86 400

C. 96 120

D. 102 360

43. In a computer game, the probability that Peter will win is 0.6. If he plays the game 4 times, find the probability that he will win exactly twice.

A. 0.3456
B. 0.4132
C. 0.4384
D. 0.5678

44. In a test, the standard deviation of the test scores is 6 marks. John scores 42 marks and his standard score is –3. If Mary's standard score is 2, then her score in the test is

A. 66 marks.
B. 72 marks.
C. 78 marks.
D. 84 marks.

45. Let m_1, q_1 and v_1 be the median, the inter-quartile range and the variance of a group of numbers $\{a_1, a_2, a_3, \ldots, a_{45}\}$ respectively. If the median, the inter-quartile range and the variance of the group of numbers $\{4 - 3a_1, 4 - 3a_2, 4 - 3a_3, \ldots, 4 - 3a_{45}\}$ are m_2, q_2 and v_2 respectively, which of the following must be true?

I. $m_2 = 4 - 3m_1$
II. $q_2 = 3q_1$
III. $v_2 = 9v_1$

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

END OF PAPER